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Knowing When and Why to Refuse: Defect-Aware Explanation and Recovery for Assistive Visual Question Answering

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ABSTRACT Visual question answering (VQA) systems for blind and low-vision users must do more than return an answer: they must know when an answer is unreliable, explain why the interaction failed, and help the user recover. This article studies a lightweight reliability layer for assistive VQA that combines answerability triage, human-labeled image-quality defect diagnosis, selective-prediction diagnostics, and actionable retake guidance. Using VizWiz-VQA and VizWiz-QualityIssues, real answerability labels are joined with seven human-annotated quality defects: blur, brightness, darkness, obstruction, framing, rotation, and unrecognizable content. Small prediction heads trained on frozen CLIP, DINOv2, and MobileNetV3 embeddings estimate answerability and defects, while frozen ViLT and BLIP VQA models provide answers and confidence scores. The strongest triage head reaches AUROC 0.7926 with CLIP; the strongest defect head reaches mAP 0.6309 with DINOv2; and CLIP obtains ungated actionable recovery 0.7196. When retake guidance is gated by refusal at a 90% target coverage, CLIP reaches gated actionable recovery 0.875 with gated false refilm rate 0.050 under the ViLT gate. Selective prediction is model-dependent: auxiliary reliability signals do not significantly improve ViLT confidence after false-discovery-rate correction, but they do improve the weaker BLIP confidence signal, with full-stack learned risk models improving BLIP AURC by up to 0.0179. Defect diagnosis is therefore best viewed not as a universal replacement for confidence-based risk ranking, but as an interpretable refusal-and-recovery mechanism whose ranking value depends on the underlying VQA model.

INDEX TERMS Accessibility, assistive technology, calibration, image quality assessment, selective prediction, visual question answering.

I. INTRODUCTION

VISUAL question answering (VQA) has become a natural interface for assistive technology: a blind or low-vision user can capture an image, ask a question, and receive a textual answer. Yet this setting is also unusually fragile. Images are captured in uncontrolled conditions, often without visual feedback, and the content needed to answer the question may be blurred, poorly framed, occluded, overexposed, underexposed, rotated, or impossible to recognize. The VizWiz benchmark was created precisely to reflect this real-world setting: its questions originate from blind users and frequently cannot be answered from the submitted image [1].

Reliability in assistive VQA is therefore not only a matter of answer accuracy. A system must decide when to abstain, but it should also explain why the image or question is not answerable and provide useful recovery advice. Existing reliable VQA work has largely framed this problem as selective prediction: answer when the model is likely correct and abstain otherwise [2], [3]. This framing is essential, but it does not directly tell a user how to fix the failed interaction. Separately, the VizWiz-QualityIssues dataset provides human annotations for real image-quality flaws in images captured by blind users [4]. These labels create an opportunity to connect abstention with explanation and recovery.

This article studies that connection. We build a lightweight reliability layer around a frozen VQA model. The layer predicts answerability and image-quality defects from frozen visual embeddings, harvests the VQA model's answer confidence, evaluates selective-prediction behavior, and maps predicted defects to retake guidance. Importantly, our experiments show a negative but useful result: defect-aware selective-prediction scores, answerability-triage scores, and learned combinations of confidence with auxiliary reliability features do not yield a positive risk-coverage improvement over the VQA model's own confidence after false-discovery-rate correction. This finding changes the role of defect diagnosis. Rather than replacing confidence as the risk-ranking signal, defect diagnosis provides interpretable refusal reasons and actionable recovery advice.

Our contributions are:

- We join VizWiz-VQA answerability labels with VizWiz-QualityIssues defect labels to study answerability, quality diagnosis, and recovery in one assistive VQA pipeline.
- We train lightweight answerability and multi-label defect heads on frozen CLIP [5], DINOv2 [6], and MobileNetV3 [7] embeddings, benchmarking modern visual backbones under identical evaluation conditions.
- We introduce the Actionable Recovery Rate (ARR) and False Refilm Rate (FRR) to evaluate whether predicted defects lead to useful retake guidance rather than merely abstention, and we report both ungated diagnostics and deployment-aligned refusal-gated rates.
- We report a selective-prediction diagnostic spanning defect-conditioned calibration, predicted-defect risk models, answerability-triage scores, confidence-plus-triage models, full-stack models, and an oracle ground-truth-defect model across two frozen VQA models. The results are model-dependent: auxiliary signals do not improve ViLT confidence after correction, but they significantly improve the weaker BLIP confidence signal.

II. RELATED WORK

A. ASSISTIVE VISUAL QUESTION ANSWERING

The VizWiz VQA dataset introduced a goal-oriented VQA benchmark whose images and spoken questions come from blind users [1], [8]. Unlike many synthetic or curated VQA datasets, VizWiz contains conversational questions, low-quality images, and a substantial number of unanswerable examples. This makes it a natural testbed for reliability and refusal in assistive VQA.

B. IMAGE QUALITY IN REAL-WORLD ASSISTIVE IMAGES

Chiu et al. introduced VizWiz-QualityIssues, a dataset of image-quality annotations for images captured by blind users [4], [9]. The labels include whether image content is recognizable and which flaws are present. Their work links image-quality assessment to practical vision tasks including captioning and VQA. Our article uses these human defect labels not only as an image-quality prediction task, but also as the basis for refusal explanation and retake guidance.

C. SELECTIVE PREDICTION AND RELIABLE VQA

Selective prediction allows a model to answer only when it is sufficiently reliable, trading coverage against risk [10]. Reliable VQA work has applied this idea to VQA systems, showing that softmax confidence can be insufficient and that learned selection functions can improve abstention behavior [2], [3]. Recent work further studies selectively answering visual questions under uncertainty [11], identifying unreliable responses from black-box vision-language models via consistency and uncertainty [12], and reducing unnecessary abstention in vision-language reasoning [13]. Our experiments differ in focus: rather than improving the selector itself, we ask what auxiliary image-quality signals add to VQA confidence, and show that their value depends on the VQA model and confidence type.

D. FROZEN VISUAL AND VISION-LANGUAGE MODELS

We evaluate frozen visual representations from CLIP [5], DINOv2 [6], and MobileNetV3 [7]. We use ViLT [14] as a frozen discriminative VQA model to harvest answer predictions and confidence scores. This design keeps the reliability layer lightweight and separates reliability evaluation from VQA model training.

III. DATA

A. DATASETS

We use VizWiz-VQA [1] and VizWiz-QualityIssues [4]. VizWiz-VQA provides an image filename, a natural-language question, ten crowdsourced answers, answer-type metadata, and an answerability flag. VizWiz-QualityIssues provides image-level quality labels: blur, bright, dark, obstruction, framing, rotation, and unrecognizable content. Each image was labeled by five crowdworkers; following the dataset authors, a label is considered present when at least two crowdworkers selected it [4].

B. JOINING LABELS

The two datasets are joined by image filename and split. We keep examples that have both VQA answerability labels and quality-issue labels. In the current run this produces 24,319 rows: 20,000 training rows and 4,319 validation rows. Validation is further split into a calibration subset and a report subset. All thresholds, temperatures, and learned risk-score models are fit on calibration data only; all reported selective-prediction metrics are computed on the report subset.

C. LABEL DISTRIBUTION

The answerable rate after joining is 0.7214. Quality labels are highly imbalanced. Framing and blur are the most common defects, with positive rates 0.5606 and 0.4161, respectively. Bright, dark, and obstruction are much rarer, with positive rates 0.0539, 0.0563, and 0.0352. These imbalances motivate per-defect reporting and caution against relying on accuracy alone. Figure 1 summarizes the label structure.

IV. METHOD

Figure 2 summarizes the pipeline. A frozen image encoder produces visual embeddings. Lightweight heads predict answerability and quality defects. Separately, a frozen VQA

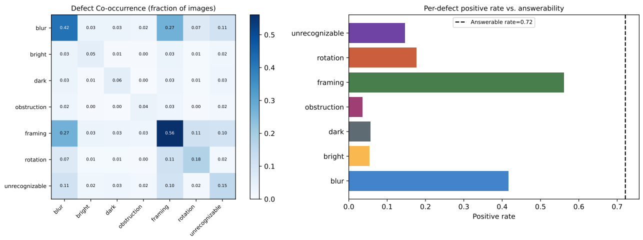

FIGURE 1. Quality-label structure in the joined data. The left panel shows pairwise defect co-occurrence, while the right panel shows per-defect prevalence relative to the answerable rate. The skewed distribution motivates per-defect metrics and recovery analysis rather than aggregate accuracy alone.

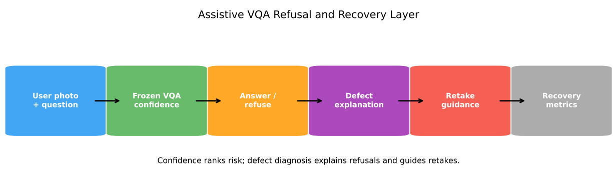

FIGURE 2. Overview of the reliability layer. Frozen visual features feed answerability and defect heads. Frozen VQA confidence provides the main risk-ranking signal. Predicted defects provide refusal explanation and retake guidance.

model answers the question and emits confidence. The system can then decide whether to answer, refuse, and provide retake guidance.

A. FROZEN BACKBONES

We compare three frozen image encoders: CLIP ViT-B/32, DINOv2 ViT-S/14, and MobileNetV3-Large. The resulting embeddings are cached once and reused by all downstream heads. This keeps the experiments reproducible and avoids recomputing expensive features.

B. ANSWERABILITY TRIAGE

For each backbone, a small multilayer perceptron predicts the VizWiz answerability label. Heads are trained on the training split with five random seeds; thresholds are selected on the calibration split, and final metrics are reported on the held-out report split as the mean and standard deviation over seeds.

C. DEFECT DIAGNOSIS

The defect head is a multi-label classifier over seven labels: blur, bright, dark, obstruction, framing, rotation, and unrecognizable. Each label is evaluated one-vs-rest, and macro/micro summaries are reported alongside per-defect AUROC and AUPRC.

D. FROZEN VQA CONFIDENCE

We use a frozen ViLT model fine-tuned for VQA to obtain a predicted answer and a maximum-softmax confidence for each example. As a robustness test, we also harvest predictions from BLIP-VQA-base, a generative model whose confidence is the length-normalized sequence probability, computed as the geometric mean of generated-token probabilities. VizWiz answer accuracy is computed as $\min(m/3, 1)$, where m is the number of exact matches between the predicted answer and the ten human answers.

E. RECOVERY MAPPING

Each predicted defect maps to a fixed user-facing retake action. For example, blur maps to holding the camera steady and refocusing, dark maps to adding light, and framing maps to stepping back so the full object is visible. This deterministic mapping is used to evaluate actionable recovery.

V. EVALUATION METRICS

A. ANSWERABILITY AND DEFECT PREDICTION

Answerability triage is evaluated using AUROC, AUPRC, F1, balanced accuracy, and a majority-class baseline. Defect diagnosis is evaluated as a multi-label problem using per-defect AUROC and AUPRC, macro-F1, micro-F1, and mean average precision (mAP).

B. SELECTIVE PREDICTION

Selective prediction is evaluated with risk-coverage curves and the area under the risk-coverage curve (AURC) [10]. Lower AURC indicates a better risk ranking. For each VQA model, we compare global VQA confidence, temperature-scaled confidence [15], defect-conditioned calibration, learned risk scores that combine confidence with predicted defects, answerability-triage scores, learned confidence-plus-triage scores, learned full-stack scores using confidence, triage, and predicted defects, and a learned risk score that combines confidence with ground-truth defects (an oracle upper bound on defect information). Statistical comparisons use paired bootstrap confidence intervals on the report split. We also apply Benjamini–Hochberg false-discovery-rate (BH-FDR) correction across each selective-prediction family of tests.

C. ACTIONABLE RECOVERY

We define the Actionable Recovery Rate (ARR) as the fraction of unanswerable examples for which the predicted defect leads to a corrective action matching a ground-truth defect. Concretely, the system selects the top-scoring predicted defect if its probability exceeds 0.5; the example counts as a recovery hit if that defect is present in the ground truth, and as a miss if no defect is predicted or the top prediction is wrong. We define the ungated False Refilm Rate (FRR) as the fraction of answerable examples for which the defect head would suggest a retake, i.e., predicts any defect above threshold. This is a conservative burden proxy: many answerable VizWiz images genuinely contain defects, so a truthful defect prediction on an answerable image still counts against FRR. For deployment, we also report refusal-gated rates at target coverages of 90%, 80%, and 70%, where retake guidance is issued only when confidence falls below a calibration-selected refusal threshold.

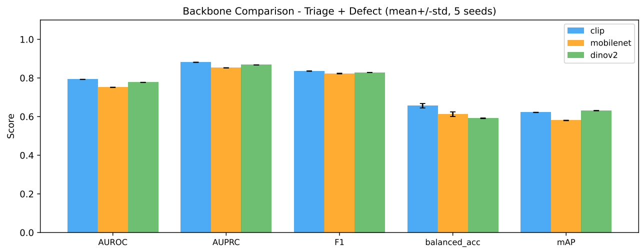

FIGURE 3. Backbone comparison for answerability triage and multi-label defect diagnosis. CLIP is strongest for answerability triage, while DINOv2 gives the highest defect mAP. Error bars show variation over five random seeds.

D. JOINT VERSUS CASCADE

To quantify architecture effects, we compare a cascade design with separate answerability and defect heads against a joint multi-task head. We report the change in answerability AUROC between joint and cascade designs.

VI. RESULTS

A. DATA SANITY

The joined dataset contains 24,319 rows, with 20,000 training examples and 4,319 validation examples. The frozen ViLT model obtains mean VizWiz answer accuracy 0.1853 and mean confidence 0.4469, confirming that the downstream reliability problem is challenging.

B. ANSWERABILITY TRIAGE

CLIP performs best for answerability triage, reaching AUROC 0.7926 ± 0.0005 and AUPRC 0.8811. DINOv2 reaches AUROC 0.7769 ± 0.0005, while MobileNetV3 reaches AUROC 0.7512 ± 0.0009. All three exceed the majority-class F1 baseline of 0.809 (model F1 0.823–0.836), although the margins are modest; balanced accuracy (0.59–0.66) gives a more conservative picture of triage quality. Table 1 and Figure 3 summarize the comparison.

C. DEFECT DIAGNOSIS

DINOv2 provides the strongest defect diagnosis by mAP, reaching 0.6309 ± 0.0009. CLIP follows at 0.6216 ± 0.0007, and MobileNetV3 reaches 0.5801 ± 0.0011. Across backbones, unrecognizable content is the easiest defect to identify (AUROC 0.93 with CLIP), while framing is the hardest by AUROC (0.83). Under AUPRC, the rare defects—obstruction, bright, and dark—drop sharply (0.33–0.48), exposing the effect of class imbalance. Figure 4 shows the per-defect breakdown.

D. ACTIONABLE RECOVERY

CLIP obtains the highest ungated actionable recovery rate, ARR 0.7196, followed by DINOv2 at 0.6907 and MobileNetV3 at 0.6278. The corresponding ungated false refilm rates are high, near 0.78. As noted in Section V, this partly reflects the genuine prevalence of defects among answerable images (framing alone affects 56% of images), and it should be interpreted as a diagnostic burden proxy rather than a deployed policy. Figure 5 breaks ungated recovery down by defect type.

When recovery is gated by the answer/refuse decision, the burden changes substantially. At a 90% target coverage under the ViLT confidence gate, CLIP reaches gated $\mathrm{ARR} = 0.875$ among refused unanswerable examples while reducing gated FRR to 0.050 on answerable examples. Under the BLIP gate, achieved coverage is 0.883 and CLIP reaches gated $\mathrm{ARR} = 0.794$ with gated $\mathrm{FRR} = 0.102$. At lower coverage, the system refuses more cases, increasing burden while reducing risk on the answered subset. The residual risk on answered examples remains high because both frozen VQA models are weak on VizWiz; the gate shifts risk in the right direction, but absolute answer quality is bounded by the underlying VQA model. Table 2 summarizes the ViLT deployment-aligned tradeoff.

E. SELECTIVE PREDICTION

The original hypothesis was that defect-aware risk scores would consistently improve selective prediction over global VQA confidence. The two-model diagnostic gives a more nuanced answer. For ViLT, global confidence is already the strongest risk-ranking signal we evaluated. It achieves AURC 0.5341 on the report split (a random ordering yields 0.7460). Temperature scaling [15] leaves the ranking unchanged and slightly worsens expected calibration error on the report split (0.194 raw versus 0.202 scaled). Defect-conditioned calibration produces small and statistically unsupported AURC changes ($p \geq 0.30$ across backbones), and learned confidence-plus-defect models are also unsupported ($p \geq 0.42$). Triage score alone is much worse than ViLT confidence for all backbones (improvement -0.176 to -0.185, $p < 0.001$), while learned confidence-plus-triage and full-stack models produce only small gains that do not survive BH-FDR correction. Across 22 ViLT selective-prediction tests from E7, E7b, and E7c, no positive improvement over global confidence remains significant at FDR 0.05.

The BLIP robustness experiment changes the interpretation. BLIP-VQA-base has report accuracy 0.2106 and a weaker global-confidence risk ranking ($\mathrm{AURC} = 0.6242$). In this setting, auxiliary reliability signals do help. With CLIP defects, a confidence-plus-predicted-defects model improves AURC by $+0.0153$, and the full stack improves AURC by $+0.0179$; both survive BH-FDR correction. With MobileNetV3 defects, the corresponding improvements are $+0.0137$ and $+0.0172$, also significant after correction. Confidence-plus-triage improves BLIP risk ranking for all three backbones ($+0.0046$ to $+0.0069$), while triage alone remains much worse than confidence. The ground-truth-defect oracle does not help BLIP (-0.0007, $p = 0.82$), suggesting that the gain comes from learned calibration interactions with predicted signals rather than from defect labels as a simple oracle ordering. Figure 6, Figure 7, Table 3, and Table 4 summarize these diagnostics.

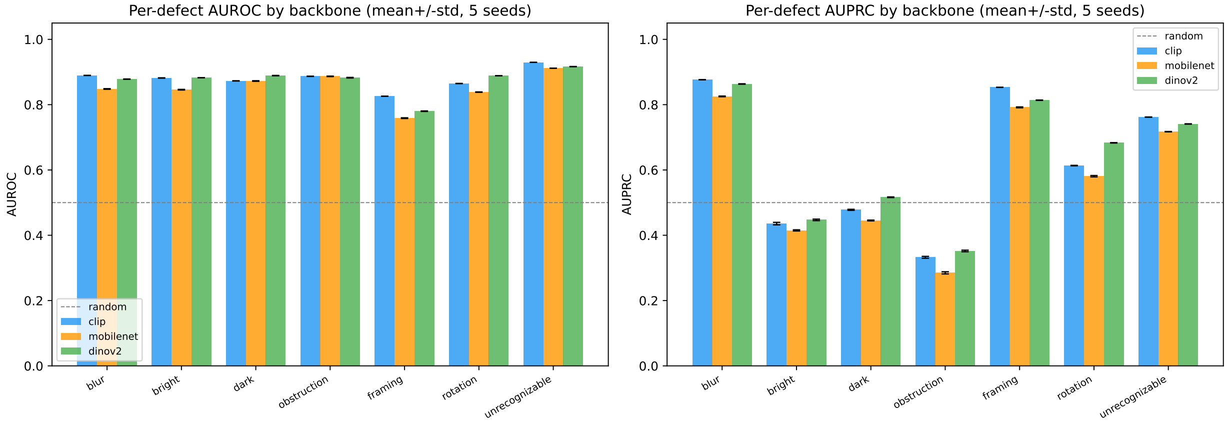

FIGURE 4. Per-defect diagnosis performance by backbone. AUROC is high for most defects, with framing the hardest label by AUROC. AUPRC exposes the effect of class imbalance: the rare defects (obstruction, bright, and dark) score far lower under AUPRC despite high AUROC.

TABLE 1. Core reliability-layer results. Triage AUROC and defect mAP are mean $\pm$ standard deviation over five seeds; $\mathrm{ARR}/\mathrm{FRR}$ show bootstrap 95% confidence intervals.

Backbone	Triage AUROC	Triage AUPRC	Defect mAP	ARR [95% CI]	FRR [95% CI]	Joint−Cascade Δ
CLIP	0.7926 ± 0.0005	0.8811	0.6216 ± 0.0007	0.7196 [0.692, 0.750]	0.7746 [0.756, 0.793]	+0.0013
DINOv2	0.7769 ± 0.0005	0.8676	0.6309 ± 0.0009	0.6907 [0.660, 0.719]	0.7775 [0.760, 0.796]	+0.0028
MobileNetV3	0.7512 ± 0.0009	0.8524	0.5801 ± 0.0011	0.6278 [0.597, 0.658]	0.7780 [0.760, 0.795]	+0.0037

TABLE 2. Refusal-gated recovery at calibration-selected confidence thresholds. Coverage and risk are shared across backbones because the refusal gate uses frozen VQA confidence; gated ARR and gated FRR depend on the defect head.

Backbone	Target coverage	Achieved coverage	Risk answered	Refusal rate	Gated ARR	Gated FRR
CLIP	90%	0.919	0.726	0.081	0.875	0.050
DINOv2	90%	0.919	0.726	0.081	0.817	0.055
MobileNetV3	90%	0.919	0.726	0.081	0.769	0.054
CLIP	80%	0.821	0.698	0.179	0.809	0.120
DINOv2	80%	0.821	0.698	0.179	0.759	0.128
MobileNetV3	80%	0.821	0.698	0.179	0.668	0.125
CLIP	70%	0.688	0.657	0.312	0.758	0.217
DINOv2	70%	0.688	0.657	0.312	0.710	0.225
MobileNetV3	70%	0.688	0.657	0.312	0.637	0.222

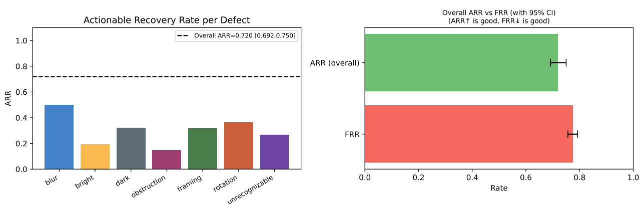

FIGURE 5. Actionable recovery from predicted defects. The left panel breaks down recovery by defect type; the right panel contrasts overall actionable recovery against the false refilm rate. The high false refilm rate underscores the need for cost-sensitive, refusal-gated recovery policies.

F. QUALITATIVE REFUSAL BEHAVIOR

Figure 8 shows a rule-sampled qualitative grid from the report split using fixed seed 7. The grid is not hand-picked: rows are sampled from high-confidence correct answers, high-confidence wrong answers, good refusals where the predicted defect matches a ground-truth defect, and wrong-reason refusals where the system refuses but diagnoses the wrong visual cause. The second row is especially important for assistive VQA: examples such as "can you read anything on this?" receive high-confidence but wrong answers, motivating a refusal layer even when a VQA model appears certain. The bottom row shows the honest failure mode of the explanation head: the system may correctly avoid answering while giving the wrong retake reason.

G. JOINT VERSUS CASCADE

The joint head gives small but consistent AUROC gains over the cascade design: +0.0013 for CLIP, +0.0028 for DINOv2, and +0.0037 for MobileNetV3. These gains are modest, suggesting that the choice of backbone matters more

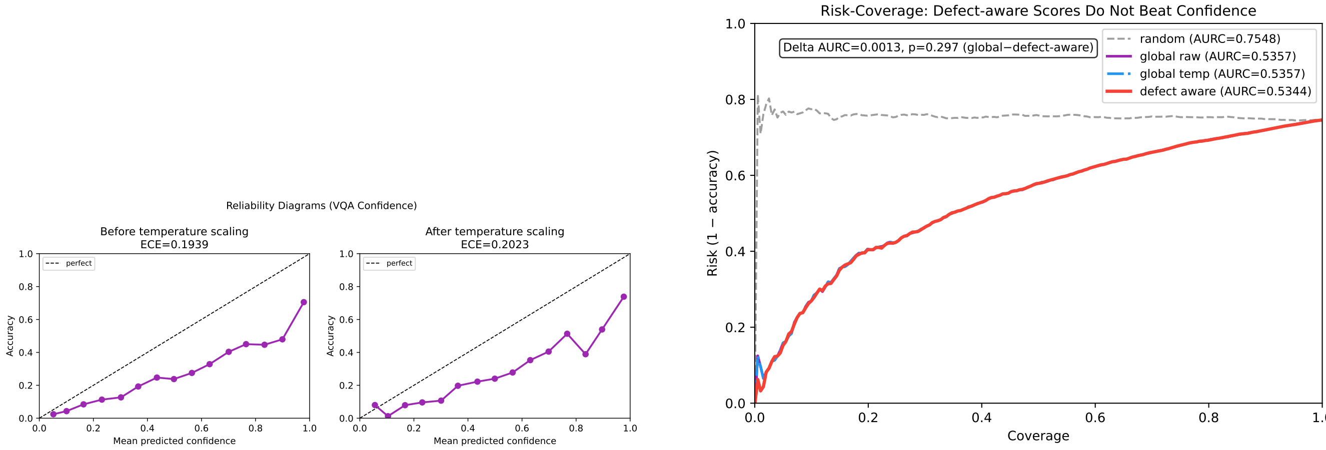

FIGURE 6. Calibration and selective-prediction behavior for frozen VQA confidence. Temperature scaling does not close the reliability gap, and defect-aware risk scores do not improve the risk-coverage curve over global confidence.

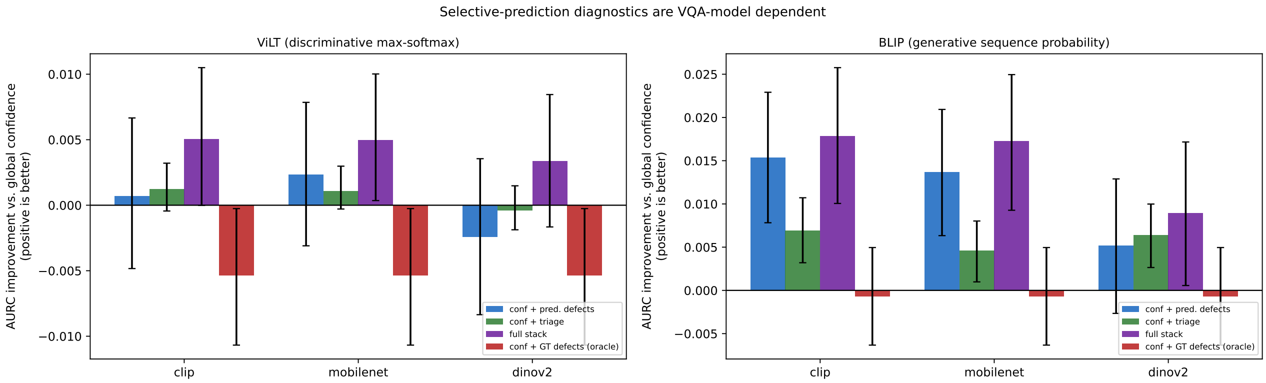

FIGURE 7. Selective-prediction diagnostics for two frozen VQA models. Positive values indicate lower AURC than each model's own global confidence. For ViLT, auxiliary signals do not produce an FDR-significant positive gain. For BLIP, learned predicted-defect and full-stack risk models improve the weaker generative confidence signal.

than the head architecture.

VII. DISCUSSION

The results suggest a separation of roles, but also show that the separation is model-dependent. For ViLT, confidence remains the most defensible signal for ranking answer risk: neither defect-derived signals nor the answerability-triage head produce a positive improvement that survives BH-FDR correction. This is plausible because a discriminative VQA model's max-softmax confidence may already encode image quality, question difficulty, answer ambiguity, and model uncertainty in a single scalar. For BLIP, however, the generative sequence-probability confidence is a weaker risk ranker, and learned combinations with triage and predicted defects significantly improve AURC. Thus, defect and triage signals should not be presented as universally useless for selective prediction; rather, their value depends on the calibration and informativeness of the base VQA confidence.

This model-dependence should not be hidden. In assistive settings, an honest reliability layer should distinguish between knowing that an answer is risky and knowing why the interaction failed. A high-confidence answer may be returned directly when confidence is well aligned with correctness. When confidence is weaker, learned auxiliary risk models may improve selective prediction. In both cases, when refusing, the defect head can explain the likely visual cause and suggest a retake action. This design avoids overstating defect labels as a universal risk-ranking replacement while preserving their user-facing value.

The gated recovery experiment clarifies how this design should be deployed. Ungated defect predictions produce high burden because many answerable images still contain real quality issues. Once retake guidance is issued only after refusal, the 90% target-coverage ViLT setting reduces CLIP gated FRR to 0.050 while preserving gated ARR 0.875; the corresponding BLIP gate is more burdensome (gated $\mathrm{FRR} = 0.102$) and less recoverable (gated $\mathrm{ARR} = 0.794$), again reflecting model-dependent confidence behavior. Asking a blind or low-vision user to retake an image still has a real usability cost, so future systems should tune coverage and recovery thresholds to balance helpful guidance against unnecessary burden.

TABLE 3. ViLT selective-prediction diagnostics on the report split. Positive improvement means lower AURC than ViLT global confidence. BH-FDR correction is applied across the 22 ViLT selective-prediction tests from E7, E7b, and E7c; no positive improvement remains significant at FDR 0.05. The oracle model uses confidence plus ground-truth defects and is therefore backbone-independent; it is reported once.

Method	AURC improvement	p	BH-FDR result
Defect-conditioned calibration (CLIP)	$+0.0009$	0.297	n.s.
Defect-conditioned calibration (DINOv2)	-0.0001	0.923	n.s.
Defect-conditioned calibration (MobileNetV3)	$+0.0008$	0.361	n.s.
Confidence + predicted defects (CLIP)	$+0.0007$	0.798	n.s.
Confidence + predicted defects (DINOv2)	-0.0024	0.448	n.s.
Confidence + predicted defects (MobileNetV3)	$+0.0024$	0.418	n.s.
Triage score alone (CLIP)	-0.1836	< 0.001	worse
Triage score alone (DINOv2)	-0.1758	< 0.001	worse
Triage score alone (MobileNetV3)	-0.1849	< 0.001	worse
Confidence + triage (CLIP)	$+0.0012$	0.168	n.s.
Confidence + triage (DINOv2)	-0.0004	0.604	n.s.
Confidence + triage (MobileNetV3)	$+0.0011$	0.140	n.s.
Confidence + triage + predicted defects (CLIP)	$+0.0051$	0.053	n.s.
Confidence + triage + predicted defects (DINOv2)	$+0.0034$	0.205	n.s.
Confidence + triage + predicted defects (MobileNetV3)	$+0.0050$	0.038	n.s.
Confidence + ground-truth defects (oracle)	-0.0053	0.039	n.s.

TABLE 4. BLIP selective-prediction diagnostics on the report split. Positive improvement means lower AURC than BLIP global confidence ($\mathrm{AURC} = 0.6242$). BH-FDR correction is applied across the 16 BLIP E7d tests; seven positive improvements remain significant at FDR 0.05.

Method	AURC improvement	p	BH-FDR result
Confidence + predicted defects (CLIP)	$+0.0153$	< 0.001	better
Confidence + predicted defects (MobileNetV3)	$+0.0137$	< 0.001	better
Confidence + predicted defects (DINOv2)	$+0.0052$	0.209	n.s.
Confidence + triage (CLIP)	$+0.0069$	< 0.001	better
Confidence + triage (MobileNetV3)	$+0.0046$	0.008	better
Confidence + triage (DINOv2)	$+0.0064$	< 0.001	better
Confidence + triage + predicted defects (CLIP)	$+0.0179$	< 0.001	better
Confidence + triage + predicted defects (MobileNetV3)	$+0.0172$	< 0.001	better
Confidence + triage + predicted defects (DINOv2)	$+0.0090$	0.043	n.s.
Triage score alone (CLIP/MobileNetV3/DINOv2)	$-0.137/-0.136/-0.130$	< 0.001	worse
Confidence + ground-truth defects (oracle)	-0.0007	0.820	n.s.

VIII. LIMITATIONS

This study uses VizWiz train and validation labels; hidden test answers are not evaluated. The frozen VQA models are ViLT and BLIP-VQA-base, both of which have low absolute accuracy on the joined data, so results may differ with stronger modern vision-language models. The selective-prediction comparison covers defect-derived signals, triage-derived signals, and their learned combinations with confidence; other auxiliary risk signals could behave differently. ARR and FRR depend on a deterministic defect-to-action map that should eventually be validated with users. The qualitative grid is rule-sampled rather than hand-picked, but it is still illustrative and not a substitute for user evaluation. Quality defects explain many image failures but not all reasons a VQA answer can be wrong, such as semantic ambiguity, text-reading failures, or missing world knowledge. Finally, spatial grounding is not included in the core results; it remains future work.

IX. CONCLUSION

We studied a defect-aware reliability layer for assistive VQA using real answerability and image-quality labels from VizWiz. Lightweight heads on frozen visual backbones can predict answerability and diagnose visual defects, and these predictions support actionable retake guidance. Selective prediction is model-dependent: auxiliary signals do not improve the stronger ViLT confidence ranking after false-discovery-rate correction, but they significantly improve the weaker BLIP sequence-probability confidence ranking. Refusal-gated recovery turns defect diagnosis into a practical user-facing mechanism: at 90% target coverage under the ViLT gate, CLIP gives gated ARR 0.875 with gated FRR 0.050. The resulting lesson is practical: use VQA confidence as the starting risk signal, learn auxiliary combinations when confidence is weak, and use defect diagnosis to explain refusal and guide recovery.

DATA AND CODE AVAILABILITY

Both datasets are publicly available from the VizWiz project [8], [9]. The full experiment pipeline (data assembly, training, calibration, selective-prediction diagnostics, recovery metrics, and figure generation), including cached-result manifests with git hashes for every reported number, is available at https://github.com/meteorboyF/VQA-paper.

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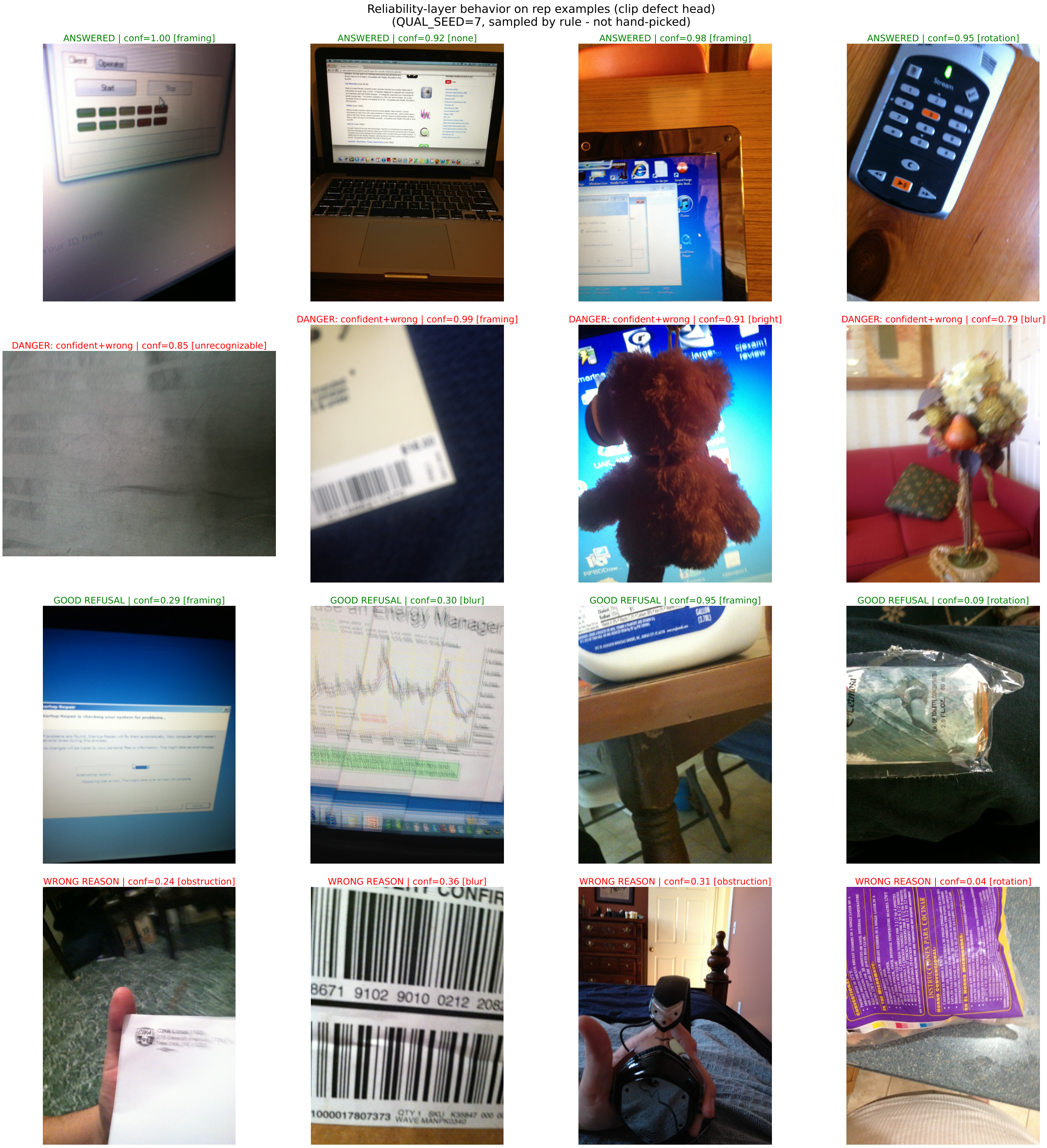

FIGURE 8. Rule-sampled qualitative examples from the report split (QUAL_SEED=7). Rows show high-confidence correct answers, high-confidence wrong answers, good refusals with matching defect-based retake guidance, and wrong-reason refusals. The figure illustrates both the motivation for refusal and the remaining challenge of explaining the correct visual failure mode.